Mathematics for Biomedical Engineering

L8:

Sequences and Series

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Key Learning Objectives

- To be able to discuss the behaviour of sequences and series that have an infinite number of terms;
- To understand the arithmetic sequence and the geometric sequence and solve problems involving these sequences;
- To be able to expand binomial expansions and solve problems involving binomial expansions;
- To be able to calculate Taylor and Maclaurin series and solve problems involving these series.

1. Introduction
   - Definitions
   - Infinite Sequences and Series
   - Arithmetic Sequences
   - Geometric Sequences

2. Pascal's Triangle and the Binomial Theorem

3. Taylor Series and Maclaurin Series

Definitions

A **sequence** is a set of numbers written down in a specific order; e.g. $-1, -2, -3, -4, -5$.

Each number in the sequence is called a **term** of the sequence.

The symbol $\dots$ is used to indicate that the sequence continues; e.g. $1, 2, 3, \dots, 20$ is the sequence of integers from 1 to 20; $1, 3, 5, 7, 9, \dots$ can be assumed to continue indefinitely.

We will denote full sequences concisely using the notation x_1 for the first term, x_2 for the second, and so on. Thus, x_k stands for the kth term. We can denote the full sequence by writing the formula to obtain the kth term; e.g.

$$x_k = 2k - 1, \qquad \text{for } k = 1, 2, 3, \dots$$

`Example:` Find a formula for the kth term of the sequence $2, 4, 6, 8, \dots$

Definitions

A **series** is formed when the terms of a sequence are added. For example, if we add the terms of the sequence $1, 3, 5, 7, 9$ we obtain the series $1 + 3 + 5 + 7 + 9$.

We use **sigma notation** to write a series concisely; e.g. the sum of the first 10 odd numbers can be written

$$\sum_{k=1}^{10} 2k - 1,$$

where the lowermost and uppermost values of k are placed below and above $\sum$, respectively.

`Example:` Write explicitly the terms of the series $\sum_{k=1}^{5} \frac{1}{k}$.

Series are important because the solutions of some mathematical problems can be expressed as series.
For example, **Taylor** and **Maclaurin** series are used to provide approximations or estimates of function values.

We continue by studying the behaviour of sequences and series that have an infinite number of terms.

The Limit of an Infinite Sequence

Finite sequences stop after a finite number of terms; e.g. $1, 3, 5, 7, 9$.

Infinite sequences continue indefinitely; e.g. $1, 3, 5, 7, 9, \ldots$

The terms of some sequences get closer and closer to a fixed value; e.g. the terms of

$$x_k = \frac{1}{k}, \qquad \text{for } k = 1, 2, 3, \ldots$$

approach to the value 0 as k increases.
We say that 'as k tends to infinity, x_k tends to zero', or 'the limit of x_k as k tends to infinity is zero.' We write this as

$$\lim_{k \to \infty} x_k = 0$$

A sequence is said to **converge** if it has a **limit** as k tends to infinity.

A sequence is said to **diverge** if it has **not a limit** as k tends to infinity.

`Example:` Explore the behaviour of the sequence $x_k = \sqrt{2 + \frac{1}{k}}$, $\;k = 1, 2, 3, \ldots,$ as $k \to \infty$.

The Sum of an Infinite Series

When the terms of an infinite sequence are added we obtain an **infinite series**.

For any infinite series, $\sum_{k=1}^{\infty} x_k$, we can form the **sequence of partial sums**, S_n,

$$S_n = \sum_{k=1}^{n} x_k, \qquad \text{for } n = 1, 2, 3, \dots$$

If the sequence S_n converges to a limit S, we say that the infinite series has sum S, or that it has converged to S.

There are **convergence criteria tests** to help decide whether or not a given series converges of diverges.

`Example:` Calculate the first four partial sums of the series $\sum_{k=0}^{\infty} \frac{1}{k!}$.

Special Cases

The Sum of the First n Positive Integers

The sum of the first n positive integers is given by

$$\sum_{k=1}^{n} k = 1 + 2 + 3 + \cdots + n = \frac{n}{2}(n+1)$$

The Sum of the Squares of the First n Positive Integers

The sum of the squares of the first n positive integers is given by

$$\sum_{k=1}^{n} k^2 = 1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n}{6}(n+1)(2n+1)$$

The Sum of the Cubes of the First n Positive Integers

The sum of the cubes of the first n positive integers is given by

$$\sum_{k=1}^{n} k^3 = 1^3 + 2^3 + 3^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2}\right)^2$$

Arithmetic Sequences

An **arithmetic sequence** is a sequence of numbers where each new term after the first is formed by adding a fixed amount, called the **common difference**, to the previous term in the sequence. For example, $3, 5, 7, 9, 11, \ldots$ is an arithmetic sequence with first term 3 and common difference of 2.

Key points

The k-**th term** of an arithmetic sequence is given by

$$x_k = a + (k-1)d, \text{ where } a \text{ is the first term and } d \text{ is the common difference.}$$

The **sum** of the first n terms of an arithmetic sequence is

$$S_n = \frac{n}{2}\left(2a + (n-1)d\right).$$

The sum of the terms of an arithmetic sequence is known as an **arithmetic series**.

`Example:` Find the sum of the first 23 terms of the arithmetic sequence $4, -3, -10$.

Geometric Sequences

A **geometric sequence** is a sequence of numbers where each new term after the first is formed by multiplying the previous term in the sequence by a fixed amount called the **common ratio**. For example, $1, 3, 9, 27, \ldots$ is a geometric sequence with first term 1 and common ratio 3.

Key points

The k-**th term** of a geometric sequence is given by

$$x_k = ar^{(k-1)}, \text{ where } a \text{ is the first term and } r \text{ is the common ratio.}$$

The **sum** of the first n terms of a geometric sequence is

$$S_n = \frac{a(1-r^n)}{1-r} \quad \text{(valid only if } r \neq 1)$$

The **sum to infinity**, S_∞, can be found if the common ratio is less than 1 in modulus:

$$S_\infty = \frac{a}{1-r} \quad \text{provided } -1 < r < 1.$$

The sum of the terms of a geometric sequence is known as a **geometric series**.

`Example:` Find the seventh term of the geometric sequence $2, -6, 18, \ldots$.

Pascal's Triangle and the Binomial Theorem

These are techniques for expanding expressions of the form $(a+b)^n$, which is called a **binomial expansion**.

When the power n is very small it is possible to multiply out the brackets; e.g. $(a+b)^2 = a^2 + 2ab + b^2$.

For larger values of n we can use **Pascal's triangle**.

For **very** large values of n Pascal's triangle is tedious and it is simpler to use the **binomial theorem**.

Pascal's Triangle

Every entry is obtained by adding the two entries on either side in the preceding row, always starting and finishing a row with a 1.

$$\begin{array}{ccccccccccc}
&&&&&1&&&&& \\
&&&&1&&1&&&& \\
&&&1&&2&&1&&& \\
&&1&&3&&3&&1&& \\
&1&&4&&6&&4&&1& \\
1&&5&&10&&10&&5&&1
\end{array}$$

Figure: Pascal's triangle.

The $(n+1)$-th row in the triangle provides the coefficients of the binomial expansion of $(a+b)^n$. Furthermore, the terms of the expansion are composed of decreasing powers of a and increasing powers of b. For example,

$$(a+b)^2 = a^2 + 2ab + b^2$$

$$(a+b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$$

`Example:` Find $(\cos x + \sin x)^3$.

Binomial Theorem

It is useful when we want to expand $(a+b)^n$ for large n.

Binomial Formula

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k,$$

where n is an integer, a, b are real numbers, and

$$\binom{n}{k} = \frac{n!}{(n-k)!k!}, \qquad n! = n \cdot (n-1) \cdot (n-2) \cdot \ \dots \ \cdot 2 \cdot 1.$$

Remember: $\binom{n}{k}$ is the number of ways k objects can be chosen from n regardless of their order; $n!$ is the factorial (the number of ways to choose an order for n elements).

Proof of the binomial formula can be obtained by induction.

`Example:` Use the binomial theorem to write down the expansion of $(1+x)^4$.

Binomial Theorem

The coefficients in the binomial formula are the same as those in Pascal's triangle:

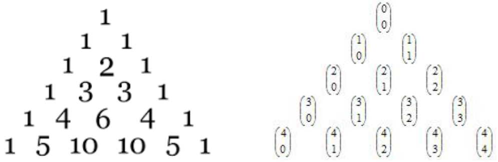

Figure: (left) Pascal's triangle and (right) binomial formula's coefficients.

Binomial Theorem When n Is Not a Positive Integer

When n **is** a positive integer, the binomial formula for $(1+x)^n$ is given by the following expression:

$$(1+x)^n = \sum_{k=0}^{n} \binom{n}{k} x^k.$$

This is a sum of a **finite number of terms**.

When n **is not** a positive integer, the binomial formula can be applied provided that x lies between -1 and 1. The expression above becomes the following **infinite series**.

Binomial Formula When n Is Not a Positive Integer

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots \qquad \text{only for } -1 < x < 1$$

`Example:` Use the binomial theorem to write down the first four terms in the expansion of $(1+x)^{1/2}$.

Taylor Series and Maclaurin Series

Suppose that the value of a function and the value of its derivatives are known at a particular point.

From this information it is possible to obtain values of the function around that point using a **Taylor series**.

The **Maclaurin series** is a special case of the Taylor series when the particular point is the origin.

Maclaurin Series

The **Maclaurin series formula** enables us to find the value of the function $f(x)$ at a point, x, close to the origin from the value of $f(0)$ and the value of the derivatives of $f(x)$ at $x = 0$; i.e. $f'(0)$, $f''(0)$, and so on.

Maclaurin Series

$$f(x) = \sum_{n=0}^{\infty} \frac{1}{n!} f^{(n)}(0) x^n = f(0) + xf'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots,$$

where $f^{(n)}(0)$ denotes the n-th derivative of f at $x = 0$.

This is an infinite series, although often we can approximate $f(x)$ by using just a finite number of terms.

It is also known as the **power series expansion** of $f(x)$.

`Example:` Obtain the Maclaurin series for $f(x) = e^x$.

Maclaurin Polynomials

In the previous example we have shown that about $x = 0$ we have

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

By taking successive partial sums of this Maclaurin series we can obtain polynomials that are approximations to e^x, known as **Maclaurin polynomials**.

For example, denoting these polynomials by $p_0(x)$, $p_1(x)$ and so on, we have

$$p_0(x) = 1, \qquad p_1(x) = 1 + x, \qquad p_2(x) = 1 + x + \frac{x^2}{2!}$$

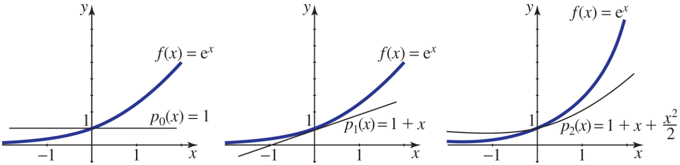

Figure: $f(x) = e^x$ together with a number of Maclaurin polynomials. As more and more terms are included in the polynomials the approximation of f improves.

Common Power Series Expansions

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \text{ for all } x$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots \text{ for all } x$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots \text{ for all } x$$

$$\log_e(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots \text{ for } -1 < x \leq 1$$

$$(1 + x)^p = 1 + px + \frac{p(p-1)}{2!}x^2 + \frac{p(p-1)(p-2)}{3!}x^3 + \cdots \text{ for } -1 < x < 1$$

$$\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \frac{17x^7}{315} + \cdots \text{ for } -\frac{\pi}{2} < x < \frac{\pi}{2}$$

$$\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \cdots \text{ for all } x$$

$$\cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \cdots \text{ for all } x$$

Note that not all the expansions are valid for any x.

Taylor Series

The Taylor series formula enables us to find the value of the function $f(x)$ at a point, $x = x_0$ (not necessary the origin) from the value of $f(x_0)$ and the value of the derivatives of $f(x)$ at $x = x_0$; i.e. $f'(x_0)$, $f''(x_0)$, and so on.

Taylor Series

$$
\begin{aligned}
f(x) &= \sum_{n=0}^{\infty} \frac{1}{n!} f^{(n)}(x_0)(x-x_0)^n \\
&= f(x_0) + (x-x_0)f'(x_0) + \frac{(x-x_0)^2}{2!} f''(x_0) + \frac{(x-x_0)^3}{3!} f'''(x_0) + \ldots,
\end{aligned}
$$

where $f^{(n)}(x_0)$ denotes the n-th derivative of f at $x = x_0$.

This is an infinite series, although often we can approximate $f(x)$ by using just a finite number of terms.

`Example:` Obtain the Taylor series expansion of $f(x) = \sqrt{x}$ about the point $x = 4$.

Taylor Polynomials

In the previous example we have shown that about $x = 4$ we have

$$\sqrt{x} = 2 + \frac{1}{4}(x-4) - \frac{1}{64}(x-4)^2 + \frac{1}{512}(x-4)^3 + \dots$$

By taking successive partial sums of this Taylor series we can obtain polynomials that are approximations to $\sqrt{x}$, known as **Taylor polynomials**.

For example, denoting these polynomials by $p_0(x)$, $p_1(x)$ and so on, we have

$$p_0(x) = 2, \qquad p_1(x) = 2 + \frac{1}{4}(x-4), \qquad p_2(x) = 2 + \frac{1}{4}(x-4) - \frac{1}{64}(x-4)^2$$

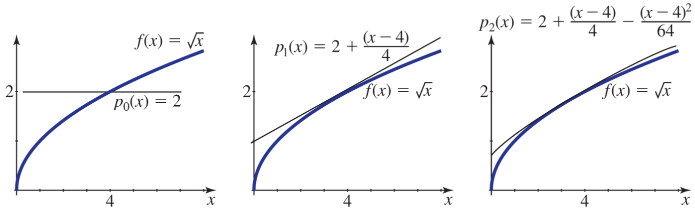

Figure: $f(x) = \sqrt{x}$ together with a number of Taylor polynomials. As more and more terms are included in the polynomials the approximation of f improves.

What We Learnt In Topic 8

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- To understand the arithmetic sequence and the geometric sequence and solve problems involving these sequences;
- To be able to expand binomial expansions and solve problems involving binomial expansions;
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